

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

Third Semester of B. Tech. (CL) Examination
November 2010

CL 204 Engineering Geology

Date: 30.11.2010, Tuesday

Time: 10:30 a.m. To 01:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Draw neat figures wherever required. Also quote suitable examples.

SECTION - I

Q - 1 What is seismic activity? Explain the types of seismic waves and their characteristics. How elastic rebound theory helps to understand earthquake? [07]

Q - 2 (a) Define Geology and Engineering Geology. Enlist various branches and allied branches of geology (atleast Ten). Write about the scope of Geology in construction field. [05]

(b) Differentiate metamorphism and diagenesis processes. Explain different types of metamorphism with suitable examples for each. [05]

(c) Define and explain briefly the following terms: [04]

(i) Colour in minerals

(ii) Specific gravity of minerals

OR

Q - 2 (a) How Igneous and Metamorphic rocks are recognized in the field? Enlist these rocks (atleast Four for each type). Describe various structures developed in igneous rocks. [05]

(b) Write short notes on: [05]

(i) Frost action and thermal effects on rocks

(ii) Carbonation and hydration process on rocks

(c) Define and explain briefly the following terms: [04]

(i) Hardness of minerals

(ii) Rocks, Minerals, Ore and Rocks forming minerals

Q - 3 (a) Give the Hatch scheme of classification of igneous rocks. How colour scheme and textures in igneous rocks are helpful to identify the rocks? [05]

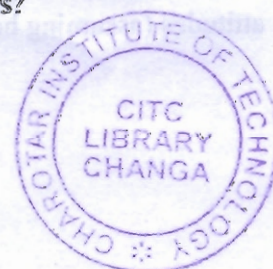
(b) (i) Write a note on various rocks as construction materials [05]

(ii) Write about grades and zones in metamorphic rocks

(c) (i) What is desert advancement? Write techniques adopted to combat this problem. [04]

(ii) How intensity and magnitude of an earthquake differs?

OR



FOR REFERENCE

- Q - 3 (a)** Give the generalized classification of sedimentary rocks. How textures in sedimentary rocks are helpful to identify these rocks? [05]
- (b)** Explain various erosional and depositional works done by underground water. Enlist various features developed due to these work. [05]
- (c)** (i) How sediments are transported by rivers? [04]
(ii) Write about Aeolian deposits and Loess.

SECTION – II

- Q - 4 (a)** Define mass movement and subsidence. Write a note on internal factors responsible for the mass movement. [05]
- (b)** Draw the sketch of an ideal fold and write various parts. Also define them. [02]
- Q - 5 (a)** What is fault in rock? Describe causative factors of fault. How they are related to the earthquake? [05]
- (b)** What is an unconformity? Describe various types of an unconformity. Write its engineering significance. [05]
- (c)** Attempt following: [04]
(i) Classification of joints
(ii) Attitudes in layered structures

OR

- Q - 5 (a)** What is fold? Write note on causative factors for the fold. How it is having influence on civil structures? [05]
- (b)** What are joints in rocks? Describe the occurrence of joints in various rocks. How they influence engineering structures? [05]
- (c)** Write the engineering consideration of faults and joints in tunnel construction. [04]
- Q - 6 (a)** Write a note on various forces affecting the dam structure. Elaborate your answer with proper sketch. [05]
- (b)** How the design of the tunnel is controlled by the geological factors? What are the geological investigations carried out for selecting tunnel alignment? [05]
- (c)** What is siltation problem in reservoir structures? State the precautionary measures and remedial measures to check the siltation problems. [04]

OR

- Q - 6 (a)** Write a note on different aquifer system for the subsurface water. Add your note with perched water, artesian water conditions and aquiclude. [05]
- (b)** Comment on the suitability of a tunnel driven through soft and hard rock formation. Also comment on the problems faced due to the tunnels driven through the area having higher water table than the tunnel alignment. [05]
- (c)** Comment on suitability of the dam structure on the layered structure with different attitudes (assuming horizontal to vertical dipping of hard rock formation). [04]
